Lab 0: SimpleScalar Setup and Orientation

Computer Architecture Laboratory

1 Purpose

This lab prepares students to build and run the SimpleScalar simulators used in the course. Students will verify the repository layout, build the simulators for PISA and Alpha, and run a small test program.

2 Repository Layout

After cloning the course repository, the repository root itself is the SimpleScalar working directory. If the repository is cloned as `SimpleScalar`, the layout is:

```
SimpleScalar/
  benchmarks/
  config/
  labs/
  scripts/
  SimpleScalar/
    experiments/
  tests-alpha/
  tests-pisa/
  sim-fast.c
  sim-safe.c
  sim-profile.c
  sim-bpred.c
  sim-outorder.c
  sim-singlecycle.c
  Makefile
```

There is no extra nested `simplesim-3.0/` directory in this repository. Run the commands in these manuals from the cloned repository root unless a lab explicitly says to enter `labs/`.

3 Simulator Overview

Simulator	Purpose
`sim-fast`	Fast functional simulation with minimal checking.
`sim-safe`	Functional simulation with additional safety checks.

Simulator	Purpose
`sim-profile`	Instruction and address profiling.
`sim-cache`	Cache hierarchy simulation.
`sim-bpred`	Branch predictor simulation.
`sim-outorder`	Detailed out-of-order processor timing simulation.
`sim-singlecycle`	Educational single-cycle timing model added for this course.

4 Part A: Build for PISA

Open a terminal in the cloned repository root:

```
cd /path/to/SimpleScalar
```

Clean any old build files:

```
make clean
```

Configure for PISA:

```
make config-pisa
```

Compile the simulators:

```
make
```

Check that simulator executables were created:

```
ls sim-fast sim-safe sim-profile sim-bpred sim-outorder sim-singlecycle
```

Run a quick PISA test:

```
./sim-safe tests-pisa/bin.little/test-math
```

5 Part B: Build for Alpha

Clean the previous build:

```
make clean
```

Configure for Alpha:

```
make config-alpha
```

Compile again:

```
make
```

Run a quick Alpha test:

```
./sim-safe tests-alpha/bin/test-math
```

6 Part C: Getting Help From a Simulator

Every simulator supports a help option:

```
./sim-profile -h
./sim-outorder -h
./sim-bpred -h
```

Use the help output to confirm the option names used in later labs.